# Binary-Field STARKs for Hyperdimensional Computing: Zero-Cost XOR Binding and Efficient Temporal Proofs

Tristan Stoltz
Luminous Dynamics
tristan.stoltz@evolvingresonantcocreationism.com

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## Abstract

Hyperdimensional Computing (HDC) represents information using high-dimensional binary vectors and manipulates them via bitwise operations—predominantly XOR binding, majority bundling, and cyclic permutation. We observe that these operations share the same algebraic substrate as binary tower field arithmetic in *Binius*, a recently proposed STARK proof system over $\mathrm{GF}(2^k)$. This alignment means that HDC's core XOR binding operation translates to *native field addition* in Binius, requiring **zero non-linear constraints**—compared to $O(N)$ constraints in prime-field STARKs, where $N$ is the vector dimensionality. We implement and benchmark a Binius proof of 16,384-bit HDC XOR binding, demonstrating **192**$\times$ **fewer constraints**, **5.7**$\times$ **faster proving**, and **2.6**$\times$ **smaller proofs** compared to prime-field STARKs. We further show that Closed-form Continuous-time (CfC) neural networks, which underpin temporal HDC processing, require only **1 AND constraint per neuron per timestep** in Binius, with all additive state updates being free. These results establish binary-field STARKs as the natural proof system for privacy-preserving HDC applications, including federated learning, anonymous governance, and decentralized identity.

**Keywords:** zero-knowledge proofs, hyperdimensional computing, binary fields, STARKs, Binius, privacy-preserving machine learning, CfC networks

## 1 Introduction

Hyperdimensional Computing (HDC) [1] encodes information as high-dimensional binary vectors (typically $D$ = 10,000–16,384 bits) and performs reasoning through three algebraically simple operations: XOR binding ($\oplus$), majority-vote bundling, and cyclic permutation ($\rho$). These operations are computationally trivial—nanoseconds on commodity hardware—yet they enable robust classification, associative memory, and temporal sequence processing [2].

A growing need exists for *privacy-preserving HDC*: proving that an HDC computation was performed correctly without revealing the underlying vectors. Applications include federated learning with verifiable gradient integrity [9], anonymous voting with provable eligibility [10], and selective credential disclosure [11]. Zero-knowledge proof (ZKP) systems provide this guarantee, but existing systems impose enormous overhead on bitwise operations.

**The Problem.** In prime-field STARK systems (e.g., Winterfell [6], StarkWare), the field's characteristic is a large prime $p$. XOR on $N$-bit values requires decomposing field elements into binary, performing per-bit XOR constraints ($c_i = a_i + b_i - 2a_ib_i$), and recomposing—yielding $O(3N)$ constraints. For 16,384-bit HDC vectors, this means $\sim$49,152 constraints per binding operation.

**Our Observation.** Binary tower fields $\mathrm{GF}(2^k)$ are constructed by successive quadratic extensions of $\mathrm{GF}(2)$. In these fields, addition *is* XOR—it is the native field operation. The

Binius proof system [4] operates over this exact algebraic structure. Therefore, HDC's XOR binding translates to field addition in Binius, requiring **zero non-linear (AND/MUL) constraints**.

**Contributions.**

1. We identify the algebraic alignment between HDC operations and binary tower fields (Section 3).

2. We implement and benchmark a Binius proof of 16,384-bit HDC XOR binding, showing 192× fewer constraints than prime-field STARKs (Section 6).

3. We demonstrate that CfC neural networks require only 1 AND constraint per neuron per timestep in Binius, enabling efficient temporal proofs (Section 4).

4. We describe the integration into Mycelix, a production Holochain-based decentralized system with 11 privacy-sensitive clusters (Section 5).

# 2 Background

## 2.1 Hyperdimensional Computing

HDC represents concepts as binary vectors $\mathbf{v} \in \{0,1\}^D$ where $D$ is large (typically 10,000–65,536). Three core operations form a complete algebra [3]:

- **Binding** ($\oplus$): Element-wise XOR. Produces a vector dissimilar to both inputs. Used for role-filler associations.

- **Bundling** ($[\cdot]$): Element-wise majority vote. Produces a vector similar to all inputs. Used for set representation.

- **Permutation** ($\rho$): Cyclic bit rotation. Encodes sequential/temporal order.

Similarity is measured by normalized Hamming distance: $\delta(\mathbf{a}, \mathbf{b}) = \frac{1}{D}\sum_{i=1}^{D} a_i \oplus b_i$.

## 2.2 STARK Proof Systems

Scalable Transparent Arguments of Knowledge (STARKs) [8] prove computational integrity without trusted setup. A computation is encoded as an Algebraic Intermediate Representation (AIR)—a set of polynomial constraints over an execution trace. The prover commits to the trace via Merkle trees, and the verifier checks random evaluation points via FRI (Fast Reed-Solomon Interactive Oracle Proof).

The field over which the AIR is defined determines the cost of different operations. In a prime field $\mathbb{F}_p$, addition and multiplication are native, but bitwise operations require expensive decomposition. In $\mathrm{GF}(2^k)$, the situation is reversed: XOR (addition) is free, but arithmetic multiplication follows the tower field's irreducible polynomial structure.

## 2.3 Binius: STARKs over Binary Tower Fields

Binius [4] is a STARK system that operates over binary tower fields $\mathrm{GF}(2^k)$ constructed via the Wiedemann tower:

$$\mathrm{GF}(2) \subset \mathrm{GF}(2^2) \subset \mathrm{GF}(2^4) \subset \cdots \subset \mathrm{GF}(2^{128})$$

The key properties relevant to HDC:

- **Addition is XOR**: $a + b = a \oplus b$ in $\mathrm{GF}(2^k)$. This is a *linear* operation requiring zero non-linear constraints.

- **AND is the fundamental non-linear gate**: Bitwise AND maps to field multiplication in GF(2), costing 1 constraint per gate.

- **5× hardware efficiency**: Binary field operations achieve 1,909,854 Mmult/s/µm$^2$ at 14nm, vs. 368,459 for Mersenne31 [4].

## 2.4 Closed-form Continuous-time Networks

CfC networks [7] are temporal neural networks with closed-form solutions:

$$h(t) = (1 - \sigma(f(t))) \cdot h(0) + \sigma(f(t)) \cdot x_\infty$$

where $\sigma$ is sigmoid and $f(t) = -\Delta t/\tau$. Unlike ODE-based Liquid Time-Constant (LTC) networks that require iterative integration (e.g., RK4 with $S \approx 16$ substeps), CfC computes each timestep in $O(1)$ operations.

# 3 HDC–Binius Algebraic Alignment

We now formalize the alignment between HDC operations and binary-field STARK constraints.

[Free XOR Binding] Let $\mathbf{a}, \mathbf{b} \in \{0,1\}^D$ be binary hypervectors. In a Binius STARK over GF($2^{64}$) with $D/64$ word-level columns, the XOR binding $\mathbf{c} = \mathbf{a} \oplus \mathbf{b}$ requires **zero AND/MUL constraints**. The binding is expressed as $D/64$ linear constraints (field additions), which are absorbed into the constraint polynomial without increasing its degree.

In GF($2^{64}$), addition is bitwise XOR on the underlying 64-bit representation. The Binius constraint system separates operations into linear (addition) and non-linear (AND/MUL). Since $\mathbf{c}_i = \mathbf{a}_i + \mathbf{b}_i$ for each 64-bit word $i$, the binding is purely linear. Linear constraints do not contribute to the AND/MUL constraint count and are verified at negligible cost during the FRI commitment phase.

[Bundling Cost] Majority-vote bundling of $n$ vectors over $D$ bits requires $O(D \cdot \log n)$ AND constraints (for the threshold comparison), plus $O(D \cdot n)$ free XOR additions for the vote counting.

[Permutation Cost] Cyclic permutation $\rho^k(\mathbf{v})$ requires zero constraints—it is a reindexing of the multilinear polynomial's variables in the Binius commitment scheme.

Table 1 summarizes the constraint costs.

# 4 CfC Temporal Proofs

The CfC closed-form step decomposes in Binius as:

Table 1: HDC Operation Constraint Costs by Proof System

| Operation | Binius | Prime STARK | |
|---|---|---|---|
| XOR binding ($D$ bits) | 0 AND | $3D$ | |
| Majority bundle ($n$ vecs) | $O(D \log n)$ | $O(D \cdot n)$ | O( |
| Cyclic permutation | 0 | $O(D)$ | |
| Hamming similarity | $O(D)$ | $O(4D)$ | |

$$\delta = x_\infty \oplus h \qquad \text{(XOR: FREE)} \qquad (1)$$
$$p = \sigma \wedge \delta \qquad \text{(AND: 1 constraint)} \qquad (2)$$
$$h' = h \oplus p \qquad \text{(XOR: FREE)} \qquad (3)$$

where $\sigma$ is provided as a witness (precomputed sigmoid value) and verified via a range check. Each CfC timestep thus costs **exactly 1 AND constraint per neuron** in Binius, with 2 free XOR operations.

For comparison, an LTC network with RK4 integration requires $S \approx 16$ substeps per timestep, each involving multiple multiplications:

$$\text{CfC: } N \cdot T \text{ ANDs} \quad \text{vs.} \quad \text{LTC+RK4: } \sim 16 \cdot N \cdot T \text{ ANDs}$$

This gives CfC a **∼16× constraint advantage** for temporal proof generation.

# 5 System Design: DASTARK

We integrate these findings into DASTARK, a dual-backend ZKP system deployed across the Mycelix decentralized platform:

- **Binius backend**: HDC-native operations (binding, bundling, CfC evolution). Zero-cost XOR, minimal AND constraints.
- **Winterfell backend**: Domain-specific AIR circuits (range proofs, threshold checks). Used where prime-field arithmetic is natural.
- **RISC0 backend**: General-purpose zkVM for complex multi-step logic. Write arbitrary Rust as the guest program.

- **Dilithium5 authentication**: Post-quantum digital signatures (CRYSTALS-Dilithium, NIST Level 5) for prover identity binding.

Backend selection is automatic based on circuit type: HDC operations route to Binius, simple comparisons to Winterfell, and complex logic to RISC0. The `ZKBackend` trait provides a uniform interface:

```
pub trait ProofBackend: Send + Sync {
    fn prove(&self, inputs: &PublicInputs,
             witness: &[f32]) -> Result<ProofResult>;
    fn verify(&self, proof: &[u8],
              inputs: &PublicInputs) -> Result<bool>;
}
```

The system is deployed across 11 Holochain clusters with unique domain tags preventing cross-protocol replay (e.g., `ZTML:Health:RecordAttest:v1`).

# 6 Evaluation

All benchmarks run on a single machine (AMD Ryzen, 32GB RAM, NixOS) with `RUSTFLAGS="-C target-cpu=native"` and release-mode compilation. Binius proofs are real cryptographic proofs generated by the Binius64 library [5] with Blake3 Merkle commitments and FRI verification. Winterfell estimates are based on the known ∼50µs/constraint proving performance of Winterfell 0.13.1.

## 6.1 HDC XOR Binding (16,384 bits)

Table 2 shows the benchmark results for proving XOR binding of two 16,384-bit binary hypervectors.

The 256 AND constraints in Binius are from structural `assert_eq` bookkeeping—the XOR operations themselves produced **zero** AND/MUL constraints. This is a **192× reduction** in non-linear constraints.

## 6.2 CfC Temporal Evolution

Table 3 shows CfC temporal proof benchmarks at two scales.

Table 2: HDC XOR Binding Proof: Binius vs. Prime-Field STARK

| Metric | Binius (GF(2)) | Prime STARK ( |
|---|---|---|
| Non-linear constraints | **256** | 4 |
| Prover time | **430 ms** | ∼2,45 |
| Verifier time | **10.4 ms** | ∼9 |
| Proof size | **75 KB** | ∼19 |

Table 3: CfC Temporal Proof in Binius (Real Proofs)

| Config | AND | Prove | Verify | Size |
|---|---|---|---|---|
| 8N × 10T | [illegible]96 | 160 ms | 12 ms | 478 KB |
| 64N × 100T | 7,360 | 410 ms | 27 ms | 156 KB |

At 64 neurons × 100 timesteps, the CfC proof requires 7,360 AND constraints with 12,800 free XOR operations. An equivalent LTC+RK4 proof would require ∼117,760 AND constraints (16× more).

## 6.3 Winterfell Range Proofs (Health)

To validate the real-proof infrastructure, we also implemented Winterfell STARK range proofs for health attestations (proving a value $\in [min, max]$ via bit decomposition). These pass 6 end-to-end tests including boundary conditions and tampered-bounds rejection.

# 7 Related Work

**ZKP for ML.** zkML [12] compiles ML inference to ZKPs with 3× proof compression. zkRNN [13] proves RNN execution but does not exploit closed-form solutions. SUMMER [14] adds recursive ZKPs for RNN training. None address HDC-specific algebraic structure.

**HDC + Privacy.** EP-HDC [15] achieves 36–1068× throughput improvement for HDC under BFV homomorphic encryption. PP-HDC [16] provides a privacy-preserving HDC inference framework. Neither considers ZKP integration.

**Binary-Field Proofs.** Binius [4] introduces STARKs over binary tower fields with 5× hardware efficiency. Binius64 [5] provides a production implementation with AND/MUL constraint separation. Our contribution is recognizing the alignment with HDC operations.

**Verifiable FHE.** vFHE [17] adds ZKP verification to FHE with 15–20% overhead. Combined with our HDC-Binius alignment, a "triple stack" (HDC + FHE + ZKP) becomes feasible where all three layers operate efficiently over binary fields.

## 8 Discussion

**The Triple Stack.** Our results suggest a unified privacy-computation-integrity stack where HDC operations are simultaneously cheap under (1) FHE (XOR-only, no multiplications—1.2× overhead vs. 10,000× for lattice FHE), (2) Binius STARKs (XOR = free), and (3) plaintext execution (bitwise ops, nanoseconds). This "triple alignment" in GF(2) is unique to HDC among representation systems.

**Limitations.** Our Winterfell comparison uses performance estimates rather than measured values for the full 49,152-constraint AIR. The Binius proofs are real. The CfC benchmark uses random sigmoid values rather than neurally computed ones; production deployment would include sigmoid lookup table constraints.

**Future Work.** (1) Implement bundling (majority vote) as a Binius circuit and benchmark. (2) Build the Winterfell HDC XOR AIR for direct measured comparison. (3) Prototype the HDC-FHE-ZKP triple stack. (4) Prove full Symthaea cognitive loop evolution (256 CfC neurons, 31 Hz, 1,000 cycles).

## 9 Conclusion

We have shown that binary-field STARKs (Binius) are the natural algebraic substrate for zero-knowledge proofs in Hyperdimensional Computing. The core HDC operation—XOR binding—requires zero non-linear constraints in Binius versus $O(3N)$ in prime-field systems, yielding 192× fewer constraints for 16,384-bit vectors. CfC temporal proofs require only 1 AND per neuron per timestep, enabling efficient verification of neural network evolution. These results remove the primary barrier to privacy-preserving HDC at scale.

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